

SPLUNK AGENTIC OPS HACKATHON · OBSERVABILITY



Your LLM got worse last night.
Splunk saw it — and rolled it back.

Agentic Ops

HEC · SPL · Dashboards

MCP Server

Hosted Models

THE PROBLEM

LLM quality degrades **silently**

A model update, a prompt change, or a drifting RAG index drops answer quality 20–50% — with **no error, no red dashboard, no alert.**

0

alerts fire

Standard monitoring watches latency and errors — never *groundedness*.

days

to detect

Teams find out from customer complaints, not telemetry.

–52%

real regression

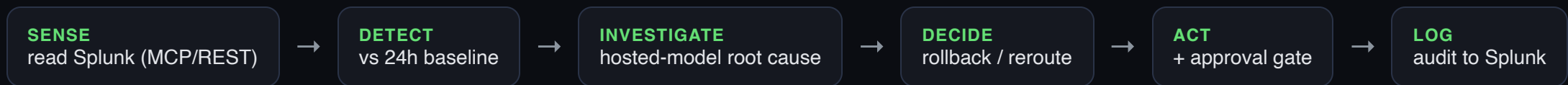
A single model bump silently halved answer quality in our live test.

The metric that matters is **never monitored**. Splunk already watches infrastructure at scale — point it at **answer quality**.

THE SOLUTION

An **agent** that runs on Splunk

LLMWatch instruments every LLM call into Splunk, then closes the loop — it doesn't just alert a human, it **remediates**.



Passive dashboards stop here

“Turn a card red, email ops-team@company.com.” The human still has to notice, investigate, and fix.

LLMWatch keeps going

Root-causes with a Splunk hosted model and rolls back the bad model — behind a human-approval gate for safety.

HOW WE USE SPLUNK

Five Splunk capabilities, one loop

CAPABILITY	HOW LLMWATCH USES IT	PRIZE
HEC	Privacy-safe ingestion of every LLM call (prompts hashed)	Core
SPL	Quality trends, regression detection, hallucination rate, cost/quality	Core
MCP Server	Agent reads Splunk securely over MCP (Cloud) — REST on local Enterprise	Best Use of MCP
Hosted Models	gpt-oss as LLM-as-judge <i>and</i> root-cause analyst	Best Use of Models
Dashboard Studio	Real-time Quality Observatory + Regression Investigation	Design

One project, eligible for the **Observability** prize + **two bonus prizes**.

PROOF — VERIFIED LIVE

Not a mock. Real Splunk.

Seeded events via HEC, ran the agent against live SPL on a Splunk Enterprise instance:

```
SENSE · pulled 2 model signals from Splunk [REST]
DETECT · gemini-2.0-flash-v2.3 dropped 52.1% (0.84 → 0.402) — REGRESSION
INVESTIGATE · gpt-oss root cause: authentication/authorization [HIGH] → rollback
ACT · executed: Active model v2.3 → v2.2. Traffic restored.
LOG · decision written to index=llmwatch (audit trail)
```

```
// read back out of Splunk — sourcetype=llm_agent_actions
{ "status":"regression_remediated", "action.kind":"rollback",
  "result.detail":"Active model v2.3 → v2.2. Traffic restored." }
```

THE QUALITY METRIC

Groundedness, scored 0–1

LLM-as-judge via a Splunk hosted model. Validated against a labelled benchmark:

WITH RETRIEVAL CONTEXT

0.621

Grounded, cites concrete specifics.

WITHOUT CONTEXT

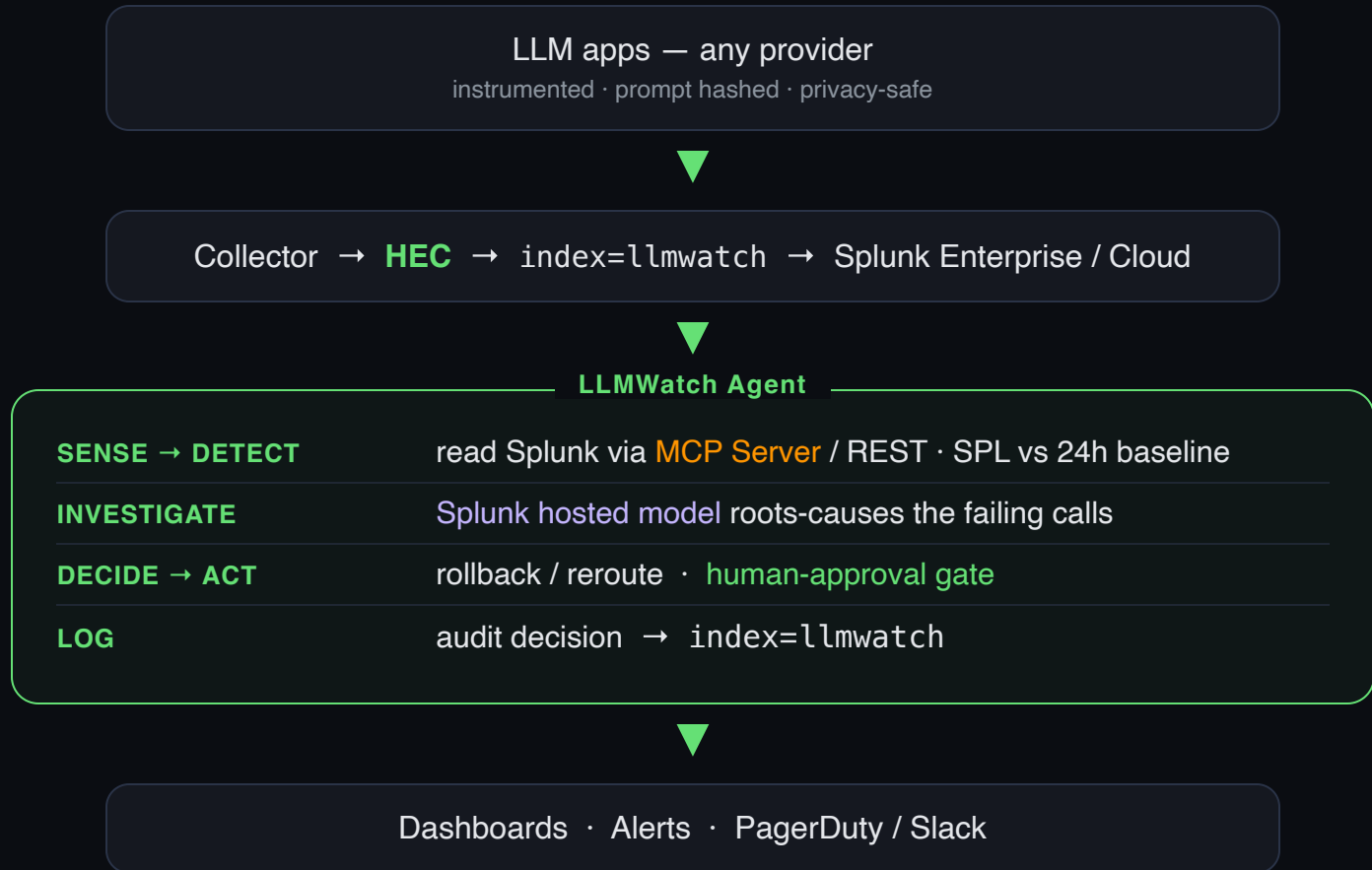
0.158

Generic, hedging, hallucination-prone.

This one score feeds **both** Splunk anomaly detection **and** the agent's detection step. **3.9× quality lift** from context — measured, not claimed.

ARCHITECTURE

HEC in · agent loop · audit out



WHY IT WINS

Judged on four criteria

Technological Implementation

Real HEC + MCP + hosted model + a working control loop. Verified live, 9 passing tests. Run it in 5 seconds.

Quality of the Idea

Meta-observability for LLMs — the groundedness metric nobody monitors. Novel and on-trend.

Potential Impact

Every team shipping LLMs has this blind spot. Catches silent drift before customers do.

Design

Enterprise dark Quality Observatory in Dashboard Studio. Demos beautifully.

Agentic, not passive — it *acts*, with responsible-autonomy safeguards.

RESPONSIBLE AUTONOMY

It acts — but it shows its work



Approval gate

Destructive actions (rollback, reroute) wait for one human click unless explicitly autonomous.



Full audit trail

Every decision logged back to Splunk — `llm_agent_actions` — queryable and reviewable.



Privacy by default

Prompts hashed before ingestion. Secrets env-only and masked in logs.

The agent earns trust the way an SRE does: **transparent decisions, reversible actions, a gate on anything destructive.**

PRODUCTION AI, OBSERVED & DEFENDED

LLMWatch

Splunk scale. LLM-as-judge groundedness. One agent closing the loop.

HEC

SPL

MCP Server

Hosted Models

Dashboard Studio

Alerts

github.com/manojmallick/llmwatch